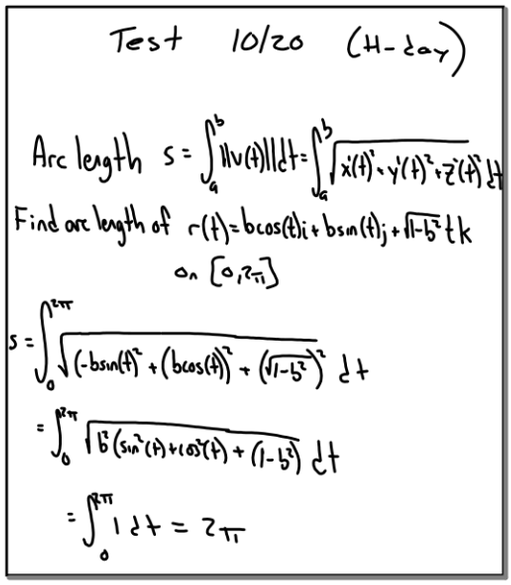
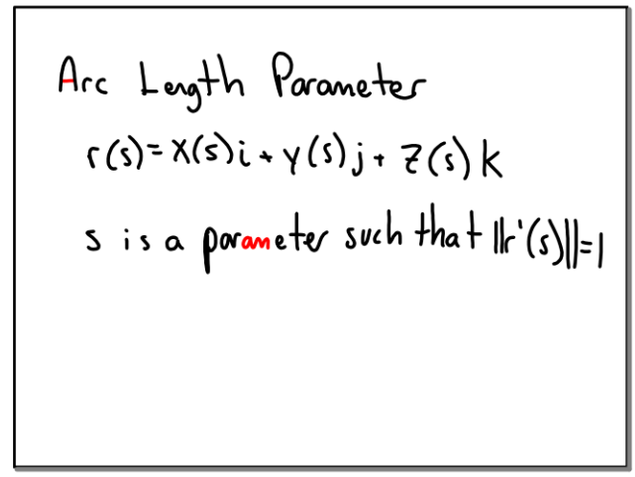
Ameet Rao's Math Notes

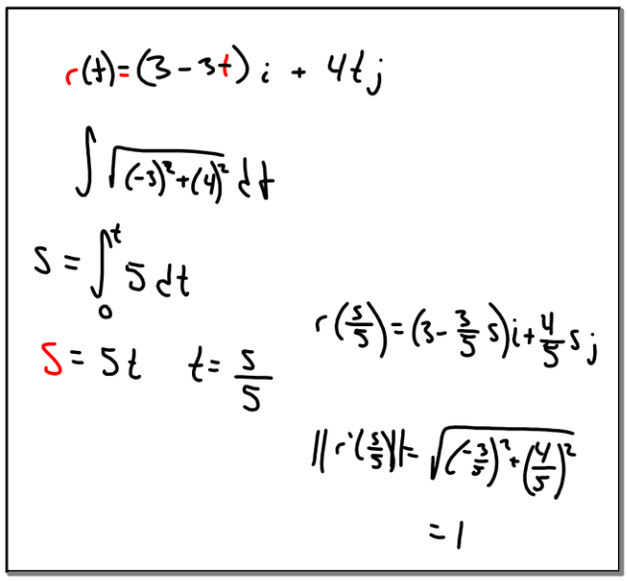
Arc Length, the Arc Length Parameter, and Curvature



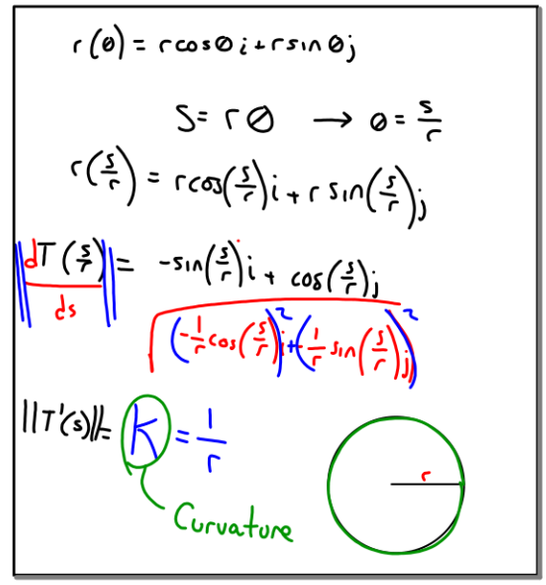
Test on 10/20. Arc length $s = \int \|r'(t)\|\, dt$. For $r = b\cos t\, i + b\sin t\, j + \sqrt{(1-b^2)}\, t\, k$ the speed is $\sqrt{(b^2 + 1 - b^2)} = 1$, so $s = \int_0^{2\pi} 1\, dt = 2\pi$.



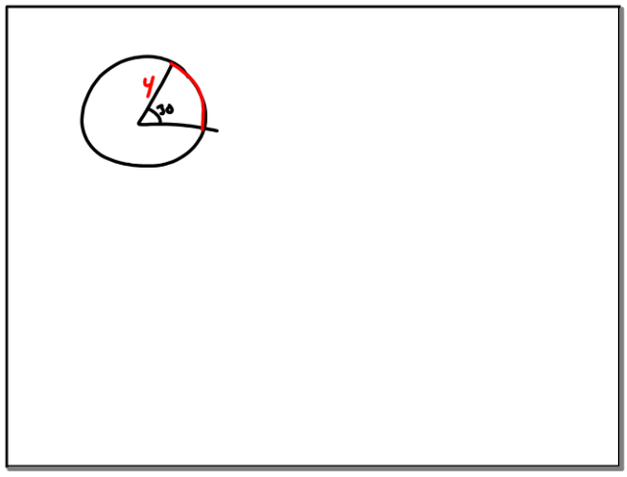
Arc length parameter: write the curve in terms of s, the distance traveled. With s as the parameter, $\|r'(s)\| = 1$, so you move at unit speed.



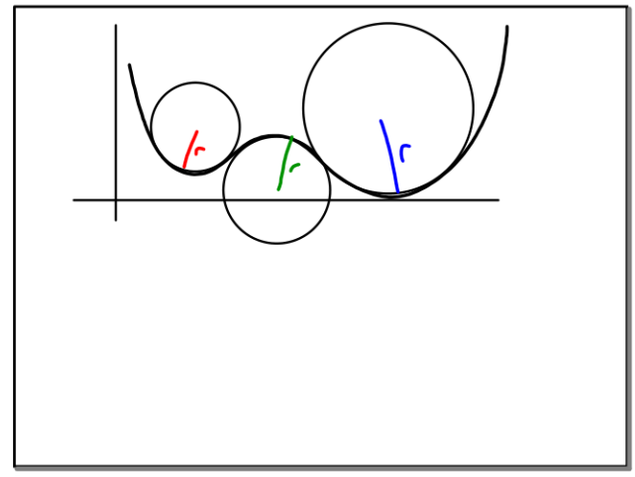
Reparametrize $r(t) = (3 - 3t)i + 4t\, j$: speed $\sqrt{(9 + 16)} = 5$, so $s = 5t$ and $t = s/5$. Then $r(s) = (3 - 3s/5)i + (4s/5)j$, and $\|r'(s)\| = 1$.



Circle of radius r: $s = r\theta$ gives $\theta = s/r$, $T(s) = -\sin(s/r)i + \cos(s/r)j$, and $\|T'(s)\| = 1/r$. That rate is the curvature K, so a circle has $K = 1/\text{radius}$.



Arc length on a circle is $s = r\theta$ with θ in radians: radius 4 and 30° ($\pi/6$ rad) give $s = 4 \cdot \pi/6 = 2\pi/3$.



Curvature measures how sharply a curve bends: fit a circle at each point. A small circle means a tight bend and large $K = 1/r$; a big circle means a gentle bend.

$$K = \frac{\|T'(t)\|}{\|r'(t)\|} = \frac{\|r'(t) \times r''(t)\|}{\|r'(t)\|^3}$$

Curvature formulas: $K = \|T'(t)\|/\|r'(t)\|$, or skip finding T and use $K = \|r'(t) \times r''(t)\|/\|r'(t)\|^3$.

$$r(t) = 2t\,i + t^2 j - \frac{1}{3}t^3 k$$

Find K

$$r'(t) = 2i + 2t\,j - t^2 k$$

$$\|r'(t)\| = t^2 + 2$$

$$K = \frac{2}{(t^2+2)^2}$$

For $r = 2t\,i + t^2\,j - (1/3)t^3\,k$: $\|r'\| = \sqrt{(4 + 4t^2 + t^4)} = t^2 + 2$ and $r' \times r'' = \langle -2t^2, 4t, 4\rangle$, length $2(t^2 + 2)$. So $K = 2/(t^2 + 2)^2$.

pg 873 - 874 - physics Application

pg 875 - All Formulas

Textbook reference: pages 873–874 cover physics applications, and page 875 lists all the formulas.

Produced in 2026